

radio frequency (RF). It died out almost as soon as it was implemented, due to the lack of a standard specification for operation, which resulted in devices from different manufacturers not being interoperable. It was soon replaced by the 802.11b standard.

802.11b

This was the first amendment to the original wireless standard introduced by the working group 11 of the IEEE 802 committee. It was released in 1999, and offered an improved theoretical data transfer rate of 11 Mbps. This standard also operated in the 2.4 GHz RF spectrum, and offers ranges of up to 8 km. The 802.11b standard was the first wireless standard to gain popularity with the masses. Since it operated on the same protocol and frequency as the original standard, manufacturers has few modifications to make in the then manufacturing technique in order to make devices 802.11b-capable. While this protocol was being developed, another task force (a group within the 802.11 working group) was developing the 802.11a protocol.

802.11a

This standard, too, was released in 1999, but products that supported it only started appearing at the end of the year 2000. The main reason for this was that 802.11a operates on the 5 GHz RF spectrum, and so manufacturing processes had to be altered, which most vendors were averse to—802.11b was catching on just fine and spinning sustainable revenue. The best thing about the 802.11a standard was the increased theoretical throughput speeds of up to 54 Mbps. 802.11a and 802.11b devices are not interoperable, unless used in conjunction with dual-mode hardware that supported both protocols. Using the 5 GHz frequency almost multiplied the theoretical throughput by a factor of five, but it also significantly reduced the range, mainly due to the additional susceptibility of 802.11a signals being absorbed by obstacles. The need arose for a standard that offered the bandwidth of 802.11a and the range of 802.11b.